

ACFS – VBS Pickup Portal Phase 1

BUSINESS REQUIREMENTS DOCUMENT V1.5

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Document Overview

1.1 Purpose

This Business Requirements Document (BRD) defines the Phase 1 scope, business processes, functional requirements, data model, assumptions, and dependencies for the ACFS Vehicle Booking System (VBS). The solution is centred on the House Bill of Lading (HBL) as the core entity throughout the logistics and booking journey.

1.2 Objectives

- Establish a HBL-centric portal that enables Logistics Service Providers (LSPs) to manage assigned HBLs, delegate shipments to other LSPs, and book pickup slots.
- Provide ACFS internal users with controlled capabilities to manage users, validate Delivery Orders (DOs), manage bookings, and perform pickup verification.
- Deliver a realistic and implementable Phase 1 scope, with explicitly identified future enhancements (fast follows) that can be added without rework of core foundations.

1.3 Intended Audience

This document is intended for:

- Project sponsor and executive stakeholders
- Product owners and business representatives
- Solution architecture and development teams
- Operations and compliance stakeholders
- Integration and infrastructure teams

1.4 Document Scope

The BRD covers:

- Actors and their responsibilities
- End-to-end user journeys for LSPs and ACFS users
- Detailed functional requirements for Phase 1
- Conceptual data model focused on HBL as the core entity
- Assumptions, constraints, and dependencies
- High-level non-functional expectations

The BRD does not define:

- Low-level technical design
- Detailed API specifications
- Detailed report layouts
- Complete non-functional specification (performance, security, DR) beyond high-level expectations

1.5 Phase 1 vs Future Enhancements

Phase 1 (this BRD) includes:

- LSP company-level login and dashboard
- HBL-centric views and actions (delegation, booking)

- DO upload and administrative validation
- Slot-based pickup bookings with fee calculation
- ACFS Admin capabilities for booking management, DO validation, pickup verification, and user management

Fast follow / future phases (explicitly out of scope for this BRD, but referenced where relevant) include:

- One-off booking party self-service flows
- On-screen configuration modules for sites and slots
- Advanced driver and vehicle management (lists, associations, separate accounts)
- Expanded reporting and analytics beyond core operational views

Actors and Responsibilities

2.1 Actor Summary

Logistics Service Provider (LSP)

- Represents wholesale freight forwarders, freight forwarders, and transport carriers collectively as a single user group.
- Access is at company account level, not individual named users.
- One login per company; internal access is managed within each company.
- Has equal feature access across all LSP entities.

ACFS Admin

- ACFS internal user with administrative privileges.
- Full access to the VBS backend portal.
- Includes at least one “Super User” instance at launch with authority to create additional users and LSP accounts.

ACFS User

- ACFS internal user with restricted permissions compared to ACFS Admin.
- Performs operational tasks within delegated permissions (for example, DO validation, pickup verification) under ACFS Admin oversight.

Maximas (External System)

- ACFS operational system of record providing shipment, HBL, milestone, customs status, and custody mapping data.
- Not a human actor but a key system participant in the end-to-end process.

2.2 Core Responsibilities by Actor

Logistics Service Provider (LSP)

- Log into the VBS portal using company credentials.
- View all HBLs assigned to the LSP, with relevant shipment and operational details.
- Select one or multiple HBLs and either:
 - Delegate shipments to another LSP, or
 - Initiate and complete a pickup booking.
- Upload DOs against HBLs where required and mark HBLs as free release where applicable.
- Provide driver and truck details as part of the booking process.

- Accept required terms and conditions for bookings and site inductions.
- Modify driver and truck details on existing bookings within permitted constraints.
- Cancel bookings within defined rules (with refunds and fee impacts handled offline by ACFS).

ACFS Admin

- Configure and maintain booking slots at site level via backend configuration (no UI in Phase 1).
- Manage bookings on behalf of LSPs, including full attribute updates and fee-free adjustments for Phase 1.
- Perform DO validation and maintain DO validation status for each HBL.
- Perform pickup verification at collection time and update booking processing status.
- Oversee HBL visibility, lifecycle states, and fail-safe manual HBL assignment.
- Manage creation, update, and deactivation of LSP accounts and ACFS internal users.

ACFS User

- Perform a subset of ACFS operational tasks (e.g. DO validation, pickup verification) subject to assigned permissions.
- Does not manage configuration or global user administration.

Maximas

- Provides HBL-to-primary-hop mapping, shipment milestones, customs status, and any required storage-related indicators to VBS.
- Acts as the system of record for shipment milestones such as Unpacked and Collected.

3. Key Concepts and Terminology (Glossary)

House Bill of Lading (HBL)

- The core entity in the solution. Represents a shipment unit around which all other data and processes are organised.

Delivery Order (DO)

- Document(s) attached to an HBL, used to authorise the release of cargo.

Logistics Service Provider (LSP)

- A company account representing wholesale freight forwarders, freight forwarders, and transport carriers, with a single shared login.

Next Hop

- The next party in the chain of custody to whom an HBL can be delegated.

Slot

- A defined time window at a given site during which pickup bookings can be made.

Cut-off

- A relative time boundary (e.g. previous working day at 16:00) after which certain actions (such as changing a slot) are no longer permitted.

High-Level Business Processes and User Journeys

4.1 LSP – HBL Management and Delegation Journey

Objective:

Allow an LSP to view assigned HBLs and delegate them to another LSP when desired.

Steps:

1. LSP Login
 - LSP accesses the VBS portal and logs in using company-level credentials.
 - On successful authentication, the system displays the LSP dashboard.
2. View Assigned HBL List
 - System displays all HBLs assigned to the logged-in LSP.
 - For each HBL, the following information is visible:
 - Lowest HBL Reference
 - Other HBL Reference
 - Container Number
 - OBL
 - Description
 - Weight
 - Volume
 - Next Hop (next step party name)
 - Milestone Status
 - Customs Status
 - Quantity
 - Pack Type
 - Storage Fee (flag)
3. Select Shipments for Action
 - User selects one or multiple HBLs from the list.
 - User chooses the action “Delegate Shipments”.
4. Search and Select Target LSP
 - User searches for another LSP by company name or email address.
 - For registered LSPs:
 - System resolves the search against existing LSP records.
 - For unregistered parties:
 - User enters only an email address.
 - System records the delegation and triggers a notification with a secure link.
 - The “one-off” style access and related flows are planned as a fast follow.
5. Attach DOs or Mark Free Release
 - For each HBL included in the delegation:
 - User can upload one or more DOs.
 - User can mark HBL as free release where applicable.
6. Confirm Delegation
 - User confirms delegation of the selected HBLs to the chosen LSP.
 - System:
 - Updates the HBL’s next-hop assignment to the delegated LSP.
 - Updates HBL lifecycle status to “Delegated”.

- Sends an email notification to the delegated LSP including:
 - Standard ACFS-provided message content.
 - A link to the portal.
- 7. Access for Recipient
 - For registered LSPs:
 - Link directs user to the login page if not already authenticated, then into a view listing the assigned HBLs.
 - For unregistered recipients (fast follow):
 - Link is secured and validated via OTP.
 - Provides temporary access to view the assigned shipments.

4.2 LSP – Pickup Booking Journey

Objective:

Allow an LSP to book a pickup slot for one or more eligible HBLs.

Steps:

1. Select HBLs for Booking
 - From the assigned HBL list, the LSP selects one or multiple HBLs to be included in a booking.
 - User chooses the action “Book Pickup”.
2. System Readiness Validation
 - For each selected HBL, the system validates:
 - Shipment milestone is “Unpacked”.
 - Shipment is fully customs cleared.
 - All required DOs have been uploaded and, where applicable, validated or marked as free release.
 - If validation fails for any HBL:
 - System clearly indicates the reason (e.g. customs not cleared, DO missing).
 - HBLs that fail readiness cannot be submitted as part of the booking.
 - Where DOs are missing but available to the LSP:
 - User can upload the DO(s) in the same flow to resolve readiness issues.
3. Calculate Load and Fees
 - System calculates total load across all selected HBLs:
 - Total Weight = sum of HBL weights.
 - Total Volume = sum of HBL volumes.
 - System calculates fees:
 - For each HBL:
 - Fee = Chargeable Weight × Rate.
 - Chargeable Weight is an HBL attribute.
 - Rate is provided and maintained by ACFS.
 - Total booking fee = Sum of all HBL fees + Minimum booking charge.
 - Minimum booking charge is provided and maintained by ACFS.
 - System displays:
 - Per-HBL calculated fee.
 - Total booking fee.
 - Total weight and volume.
4. Slot Selection

- System shows available slots for the relevant site(s).
 - For each slot:
 - Start and end times.
 - Day(s) of the week.
 - A visual indication of how busy the slot is (e.g. heat map colour) without revealing the number of existing bookings.
 - LSP selects a single slot for the booking.
5. Enter Driver and Truck Details
- User enters driver details:
 - Driver Name
 - Licence Number
 - Site Induction (Yes/No; field usage to be confirmed by ACFS)
 - User enters truck details:
 - Truck Registration (Rego)
 - Where previously entered driver and truck data will not be maintained for Phase 1. This will be introduced in the fast follow/next phase.
6. Accept Terms and Conditions
- User is presented with:
 - Booking Terms and Conditions.
 - Driver Site Induction Terms and Conditions (if required).
 - User must explicitly and individually accept:
 - Booking Terms and Conditions.
 - Driver Site Induction Terms and Conditions (if Site Induction = No; if Site Induction = Yes, driver site induction T&Cs acceptance may not be required, subject to ACFS confirmation).
 - System records acceptance flags and timestamps.
7. Payment
- User proceeds to payment.
 - System:
 - Redirects or integrates with the confirmed payment gateway (Stripe or an alternative such as Compay; final selection to be confirmed by ACFS).
 - On successful payment:
 - Confirms the booking within VBS.
 - Generates a unique booking reference.
 - Associates selected HBLs to the booking with their respective fee components.
8. Booking Confirmation
- System sends booking confirmation email to the LSP, including:
 - Booking reference.
 - Slot details.
 - Site details.
 - Summary of included HBLs and key data as appropriate.
 - LSP communicates booking details and reference to the driver offline, outside the scope of VBS.

4.3 LSP – Modify Existing Booking Journey

Objective:

Allow the LSP to modify certain aspects of an existing booking under defined conditions.

Steps:

1. Locate Booking
 - LSP searches for an existing booking by booking reference or via a bookings list view.
2. Modifications Before Change Cut-off
 - Before the defined change cut-off time for the booking's slot:
 - LSP may modify:
 - Driver details.
 - Truck details.
 - Selected slot (subject to new slot availability).
 - System re-validates slot capacity and confirms the updated booking.
 - Fee-impacting changes are not supported via the LSP interface in Phase 1; they require ACFS Admin intervention.
3. Modifications After Change Cut-off and Before Collection
 - After the change cut-off and until all HBLs in the booking are collected:
 - LSP may modify only:
 - Driver details.
 - Truck details.
 - Slot selection and any fee-impacting attributes are locked.
4. After Collection
 - Once all HBLs in the booking have reached "Collected" status:
 - Booking is effectively immutable.
 - Any required corrections are handled manually by ACFS outside the system booking flow.
5. Escalated Changes
 - For any modifications not permitted in the LSP interface (for example, fee-impacting changes), LSP must contact ACFS.
 - ACFS Admin can make such changes on the LSP's behalf in the backend.

4.4 ACFS Admin – Slot Configuration Journey (Backend-Only for Phase 1)

Objective:

Allow ACFS to configure and maintain slot definitions that will be used for bookings.

Steps:

1. Site Data Load
 - Site data (site identifiers, names, and relevant attributes) is loaded once directly into the backend as an initial import.
2. Slot Definition
 - For each site, ACFS Admin (or ACFS IT on behalf of Admin) defines one or more slots per day.
 - Slot attributes include:
 - Days of the week to which the slot applies.

- Multiple slots per day as required.
 - Start time for each slot.
 - End time for each slot.
 - Booking cut-off (relative day and time, e.g. previous working day at a specified time).
 - Change cut-off (relative day and time).
 - Heatmap indicator thresholds (for example, configuration of when a slot should display as Red, Amber, or Green in utilisation).
3. Ongoing Maintenance
- Changes to slots in Phase 1 are made by ACFS Admin or ACFS IT directly in the database/backend until a slot configuration user interface is delivered in a future phase.

4.5 ACFS Admin – DO Validation Journey

Objective:

Allow ACFS to validate DO documents attached to HBLs.

Steps:

1. Locate HBL
 - ACFS Admin or authorised ACFS User searches for HBLs using available search criteria.
 - User opens the HBL details view.
2. View DOs
 - System displays DO attachments associated with the HBL.
 - User can open each DO document.
3. Validate DOs
 - For each DO:
 - User marks the DO as “Valid” or “Invalid”.
 - System records status, validator, and timestamp.
4. HBL DO Validation Status
 - When all DOs for an HBL are marked “Valid”:
 - System sets an HBL-level “DOs Fully Validated” flag.
 - If any DO is marked “Invalid”:
 - System sends a notification to the booking party requesting upload of a corrected DO.
 - Once corrected DOs are uploaded and validated, the HBL-level flag is set to fully validated.

4.6 ACFS Admin – Pickup Verification Journey

Objective:

Allow ACFS to verify bookings and ensure conditions are met at pickup.

Steps:

1. Search for Booking
 - User searches for bookings using:
 - Booking reference, or
 - Driver name, or
 - Licence number, or
 - Truck registration.

2. Review Booking Details
 - System displays:
 - Booking details (slot, site, LSP, etc.).
 - Included HBLs.
 - DO validation status per HBL.
3. Validate Outstanding DOs
 - If DOs are still pending validation, user can:
 - Open DOs.
 - Perform DO validation as per the DO validation journey.
4. Mark Booking as Processed
 - Once all conditions are satisfied (e.g. DOs validated), user marks booking as “Processed”.
 - System updates booking status accordingly.
5. Update via Maximas Sync
 - When the shipment milestone in Maximas changes to “Collected”:
 - On next data sync, VBS updates HBL status to “Collected”.
 - When all HBLs within a booking have “Collected” status:
 - Booking status changes to “Complete”.

4.7 ACFS Admin – HBL Visibility and Manual Assignment Journey

Objective:

Provide ACFS with visibility over HBL lifecycle and the ability to perform fail-safe manual assignment.

HBL Lifecycle States:

- Unassigned – Shipment details are available but not yet assigned to any LSP.
- Assigned – Shipment has been assigned to a primary LSP (the first party in the chain of custody).
- Delegated – Shipment has been delegated from one LSP to another at any point in the chain of custody.
- Booked – HBL has been included in an active booking and has a booking reference associated with it.
- Collected – HBL’s shipment has been collected from the warehouse, based on the “Collected” milestone from Maximas.

Steps:

1. View HBL Lifecycle
 - Admin views HBLs and their lifecycle states.
 - Admin can filter or search by state as needed.
2. Fail-safe Manual Assignment
 - For Unassigned HBLs where automatic assignment has failed or is not possible, Admin:
 - Views the list of Unassigned HBLs.
 - Selects an HBL and assigns it to a specific LSP as primary custody.
 - System:
 - Updates HBL state to “Assigned”.
 - Updates next-hop LSP information.
 - Sends a notification to the newly assigned LSP.
 - Manual primary assignment does not require DOs.

4.8 ACFS Admin – User Management Journey

Objective:

Allow ACFS Admin to manage access for LSPs and ACFS internal users.

Steps:

1. Create LSP Account
 - Admin creates a new LSP company account, including:
 - LSP Name
 - Primary email address
 - Other required company details
 - System:
 - Creates the LSP record with a single login credential.
 - Sends a welcome/activation link to the specified email address.
 - Welcome links expire after 72 hours; reactivation after expiry requires ACFS support.
2. Create ACFS Users
 - Admin creates ACFS Admin and ACFS User accounts as needed.
 - ACFS Users access the portal via the existing ACFS SSO mechanism.
3. Update or Deactivate Users
 - Admin can:
 - Update user details and roles (ACFS Admin, ACFS User).
 - Deactivate or reactivate LSP accounts and ACFS user accounts as required.

Functional Requirements

5.1 LSP Functional Requirements

FR-LSP-01

System shall authenticate LSP access using a single company-level login per LSP and present an LSP dashboard upon successful authentication.

FR-LSP-02

System shall display a list of all HBLs assigned to the logged-in LSP, showing at least:

- Lowest HBL Reference
- Other HBL Reference
- Container Number
- OBL
- Description
- Weight
- Volume
- Next Hop (Next step party name)
- Milestone Status
- Customs Status
- Quantity
- Pack Type
- Storage Fee flag

FR-LSP-03

System shall allow the LSP to select one or multiple HBLs from the list and choose an action: Delegate Shipments or Book Pickup.

FR-LSP-04

For the delegation flow, system shall allow the LSP to search for a target LSP by company name or email address.

FR-LSP-05

For unregistered email addresses, system shall record the delegation and send a notification email with a secured link to the specified address; the secure temporary access flow may be delivered as a fast follow.

FR-LSP-06

System shall allow the LSP to upload one or more DO documents against each HBL and to mark HBLs as free release where appropriate.

FR-LSP-07

System shall send delegation notification emails to the target LSP, using message copy provided by ACFS and including a link to the portal.

FR-LSP-08

For the booking flow, system shall validate that each selected HBL meets readiness criteria:

- Shipment milestone is Unpacked.
 - Shipment is fully customs cleared.
 - All required DOs are uploaded and validated or marked as free release.
- If any HBL does not meet criteria, system shall prevent booking submission for that HBL and display an explanation.

FR-LSP-09

System shall allow the LSP to upload missing DOs within the booking flow to resolve readiness issues.

FR-LSP-10

System shall calculate total weight and total volume across all selected HBLs and display these to the LSP.

FR-LSP-11

System shall calculate per-HBL fees using:

- Per-HBL Chargeable Weight attribute.
 - Configured Rate (provided by ACFS).
- Fee per HBL = Chargeable Weight × Rate.

FR-LSP-12

System shall calculate total booking fee as:

Total Booking Fee = Sum of all HBL fees + Minimum Booking Charge,
where Minimum Booking Charge is provided by ACFS.

FR-LSP-13

System shall display per-HBL fee and total booking fee to the LSP before payment.

FR-LSP-14

System shall display available slots for the relevant site(s), including start time, end time, and a visual indication of slot busyness (for example, heat map), without revealing actual booking counts.

FR-LSP-15

System shall allow the LSP to select exactly one slot for each booking.

FR-LSP-16

System shall allow the LSP to enter and update driver details (Driver Name, Licence Number, Site Induction flag) and truck details (Truck Registration) for the booking.

FR-LSP-17

System shall require explicit acceptance of:

- Booking Terms and Conditions.
- Driver Site Induction Terms and Conditions (except when Site Induction = Yes, subject to ACFS confirmation).

System shall store acceptance status and timestamps.

FR-LSP-18

System shall integrate with the selected payment gateway to process booking payments and shall only confirm bookings after payment is successfully completed.

FR-LSP-19

Upon successful payment, system shall:

- Confirm the booking in VBS.
- Generate a unique booking reference.
- Associate selected HBLs with the booking.
- Send a booking confirmation email to the LSP with booking reference and key details.

FR-LSP-20

System shall allow LSPs to search for and view existing bookings, including their status and associated HBLs.

FR-LSP-21

Before the configured change cut-off, system shall allow LSPs to modify:

- Driver details,
 - Truck details,
 - Slot selection (subject to availability),
- while preventing any automated fee recalculation that affects total booking fees.

FR-LSP-22

After change cut-off and before all HBLs in the booking are collected, system shall allow LSPs to modify only driver and truck details.

FR-LSP-23

After all HBLs in a booking are collected, system shall prevent further modifications by LSP.

FR-LSP-24

System shall allow LSPs to request booking cancellations, and upon cancellation, HBLs in that booking shall become available for rebooking; financial refunds or adjustments will be handled outside VBS.

5.2 ACFS Admin and ACFS User Functional Requirements

FR-ADM-01

System shall support the storage of site data and slot definitions, including:

- Site identifiers and names,
- Slot days of the week,
- Slot start and end times,
- Booking cut-off (relative day and time),
- Change cut-off (relative day and time),

- Heatmap indicator thresholds.

FR-ADM-02

In Phase 1, slot and site definitions shall be managed via backend/database configuration rather than a front-end configuration module.

FR-ADM-03

System shall allow ACFS Admin users to manage bookings on behalf of LSPs, including updating all booking attributes, with changes in Phase 1 being fee-free for the LSP.

FR-ADM-04

System shall provide DO validation capability for ACFS Admin and authorised ACFS Users, including:

- Viewing DO attachments per HBL,
- Marking each DO as Valid or Invalid,
- Recording validator identity and timestamp.

FR-ADM-05

System shall maintain an HBL-level “DOs Fully Validated” flag that is set when all associated DOs are marked Valid.

FR-ADM-06

When a DO is marked Invalid, system shall notify the relevant booking party requesting upload of corrected DOs.

FR-ADM-07

System shall allow ACFS Admin and authorised ACFS Users to perform pickup verification by:

- Searching bookings by multiple keys (booking reference, driver name, licence number, truck registration),
- Viewing booking details, included HBLs, and their DO validation status.

FR-ADM-08

System shall allow ACFS Admin and authorised ACFS Users to mark a booking as “Processed” once all conditions (including DO validation) are met.

FR-ADM-09

System shall update HBL status to “Collected” based on Maximas milestones received via integration.

FR-ADM-10

When all HBLs within a booking have status “Collected”, system shall update the booking status to “Complete”.

FR-ADM-11

System shall maintain and display HBL lifecycle states:

- Unassigned,
- Assigned,
- Delegated,
- Booked,
- Collected.

FR-ADM-12

System shall provide a function for fail-safe manual assignment of Unassigned HBLs to an LSP as the primary custody party, without requiring DOs.

FR-ADM-13

System shall allow ACFS Admin to:

- Create LSP company accounts with single logins,

- Create ACFS Admin and ACFS User accounts,
- Update user details and roles,
- Deactivate/reactivate LSP and ACFS user accounts.

FR-ADM-14

System shall generate and send welcome or activation links for new LSP accounts, with links expiring after 72 hours.

FR-ADM-15

System shall integrate with ACFS SSO for internal ACFS Admin and ACFS User authentication.

Data Model (Conceptual)

6.1 Core Entities

HLB (House Bill of Lading)

- Attributes:
 - HBL ID
 - Lowest HBL Ref
 - Other HBL Ref
 - Container Number
 - OBL (Ocean Bill of Lading)
 - Description
 - Weight
 - Volume
 - Quantity
 - Pack Type
 - Assigned LSP
 - Next Hop LSP
 - Milestone Status (from Maximas)
 - Customs Status (from Maximas)
 - Storage Fee Flag (from Maximas)
 - HBL Lifecycle Status (Unassigned, Assigned, Delegated, Booked, Collected)
 - DOs Fully Validated Flag
 - Related Booking ID(s)

DO (Delivery Order)

- Attributes:
 - DO ID
 - HBL ID
 - File Reference / Storage Location
 - Valid/Invalid Flag
 - Validation Timestamp
 - Validator User ID

Booking

- Attributes:
 - Booking ID
 - Booking Reference

- LSP (booking party) ID
- Site ID
- Slot ID
- Booking Status (e.g. Open, Processed, Complete, Cancelled)
- Total Weight
- Total Volume
- Total Fee
- Minimum Charge Applied Flag
- Created Timestamp
- Last Modified Timestamp

Booking–HBL Link

- Attributes:
 - Booking ID
 - HBL ID
 - Chargeable Weight
 - Rate
 - Per-HBL Fee

LSP

- Attributes:
 - LSP ID
 - LSP Name
 - Primary Email
 - Company Details
 - Active Flag

User

- Attributes:
 - User ID
 - Username or Identifier
 - Role (LSP, ACFS Admin, ACFS User)
 - Linked LSP ID (for LSP users) or Internal Flag
 - Status (Active/Inactive)

Slot

- Attributes:
 - Slot ID
 - Site ID
 - Days of Week
 - Start Time
 - End Time
 - Booking Cut-off (relative definition)
 - Change Cut-off (relative definition)
 - Heatmap Thresholds

Site

- Attributes:
 - Site ID
 - Site Name
 - Location (optional)
 - Status

Payment

- Attributes:
 - Payment ID
 - Booking ID
 - Amount
 - Payment Gateway
 - Payment Status
 - Payment Timestamp

6.2 Integration Data from Maximas

Data elements provided by Maximas to VBS include:

- HBL-to-primary-hop mapping (primary LSP assignment).
- Shipment milestone statuses (e.g. Unpacked, Collected).
- Detailed customs status.
- Storage fee indicator based on LFD or related logic.

Assumptions, Constraints, and Dependencies

7.1 Assumptions

User Roles and Access

- A single, generic LSP role is used to represent wholesale freight forwarders, freight forwarders, and transport carriers.
- LSP access is provisioned at company level with one login per company. Internal distribution and control of access are managed by the company itself.
- Supported user types are: LSP, ACFS Admin, and ACFS User. One ACFS Admin will act as a “Super User” at launch with authority to create subsequent users.

Onboarding and Authentication

- There is no self-registration for Day 1. LSP and ACFS user accounts are created by ACFS Admin.
- Welcome or activation links for LSP accounts expire after 72 hours; reactivation requires ACFS support.
- ACFS internal users authenticate via the existing ACFS SSO identity platform.

Booking Eligibility and Readiness

- A booking can only be submitted when, for all included HBLs:
 - The shipment milestone is Unpacked,
 - The shipment is fully customs cleared, and
 - All required DOs are uploaded and validated or marked as free release.
- Missing or invalid DOs are resolved by upload or re-upload; a new booking is not required solely due to DO issues.
- DO sourcing (obtaining correct documents) is handled offline and is outside the scope of VBS.

Shipment and HBL Visibility

- HBL-to-primary-hop mapping and detailed customs status are available in Maximas and will be provided to VBS as ready-to-consume fields.
- Only the “next-hop” party for an HBL is displayed at a time.

- Manual HBL assignment is available only for the primary point of custody as a fail-safe mechanism and does not require DOs.

One-off Bookings

- One-off bookings, where only an email address is required and no account is created, are recognised but will not be implemented in Phase 1 and will be considered for a fast follow.

Fees and Charges

- Storage fee payment and storage fee calculation are outside the scope of VBS.
- A storage fee indicator flag will be available based on LFD; it indicates potential outstanding fees but does not drive calculation.
- A minimum charge applies per booking, not per HBL.
- A chargeable weight field is used to calculate individual HBL fees.
- Total booking fees equal the sum of all HBL fees plus the minimum booking charge.
- Total load is calculated using the total weight and volume across all HBLs.

Slot Booking and Configuration

- Each booking supports a single slot selection.
- No Day 1 restrictions are enforced on load or truck type.
- Site data will be uploaded once directly into the backend; no site configuration module is required for Phase 1.
- Heatmap indicators are informational only and do not prevent bookings.

Driver and Truck Management

- Driver details are captured and can be reused under the same LSP company account, subject to design.
- Driver data is not shared across LSPs.
- Drivers do not receive system-generated emails; booking references and instructions are provided offline by the LSP.

Booking Modifications and Cancellations

- Before the change cut-off, LSPs may update driver details, truck details, and slot selection.
- After the change cut-off and before collection, LSPs may only update driver and truck details.
- Fee-impacting changes are not supported via LSP self-service in Phase 1 and require ACFS Admin intervention.
- Booking cancellations by LSPs are permitted; any financial adjustments or refunds are managed outside VBS.

Delivery Order Validation

- DO validation is a manual administrative process performed by ACFS.
- Invalid DOs must be corrected and re-uploaded before processing continues.
- Validation criteria and decisions are governed by ACFS Operations and Compliance.
- Once all HBLs in a booking are reviewed and conditions are met, bookings are marked processed, and shipment statuses update from Maximas on the next sync cycle.

7.2 Constraints

- Phase 1 shall not include UI-based modules for slot or site configuration; these will be data-driven and maintained via backend processes.
- One-off parties and temporary access flows are not to be built for Day 1 and will not be relied upon for initial launch.
- Detailed performance and security requirements will be defined in separate technical documents; only high-level expectations are captured here.

7.3 Dependencies and Ownership

Identity and SSO

- Dependency: Integration with ACFS SSO platform for internal user authentication.
- Owner: ACFS IT / Security.

Maximas Data Feed

- Dependency: Reliable provision of HBL data, milestones (e.g. Unpacked, Collected), customs status, and storage flags.
- Owner: ACFS IT / Maximas Product Owner.

Payment Gateway

- Dependency: Selection, configuration, and commercial enablement of Stripe or an alternative gateway (e.g. Compay).
- Owner: ACFS Finance / IT.

Charging Configuration

- Dependency: Definition and maintenance of pricing parameters (minimum booking charge, rates, and chargeable weight rules).
- Owner: ACFS Finance / Product.

Communication Content

- Dependency: Provision of email templates and message copy for:
 - Delegation notifications,
 - Booking confirmations,
 - DO invalid notifications,
 - User activation/onboarding.
- Owner: ACFS Operations / Legal / Marketing (as applicable).

Slot and Site Configuration

- Dependency: Initial and ongoing slot and site configuration in backend.
- Owner: ACFS IT / Operations.

DO Validation Rules

- Dependency: Business rules and criteria for what constitutes a valid DO and how non-compliance is treated.
- Owner: ACFS Operations / Compliance.

High-Level Non-Functional Expectations

(Note: Detailed NFRs will be defined in technical documentation; the following are guiding expectations for Phase 1.)

Performance

- The system should provide responsive HBL list and search operations suitable for day-to-day operational use.
- Slot availability views and booking submission should complete within acceptable operational timeframes.

Availability

- The system should be available during ACFS-defined business operating hours for LSP and ACFS users, with specific targets defined in technical and support SLAs.

Audit and Logging

- Key actions must be auditable, including:
 - Delegation of HBLs,
 - Creation and modification of bookings,
 - DO validation decisions,
 - User creation and role changes.
- Audit logs should capture actor, timestamp, and key data changes to support compliance and operational investigations.

Security

- Access control must enforce role-based permissions for LSP, ACFS Admin, and ACFS User roles.
- Data exchange with Maximas and the payment gateway must follow ACFS security guidelines and integration standards.

This completes the BRD content for Phase 1 of the ACFS VBS HBL-centric solution and can be transferred directly into the standard BRD template.